

1 Scope

This sectional specification covers multi-element cables suitable for connecting instruments and control systems for analogue or digital signal transmission. They may or may not be screened and optionally may incorporate armouring and/or moisture or environmental protection layers.

The cables shall have a mechanically robust construction and electrical transmission handling properties. The electrical, mechanical, transmission and environmental performance characteristics of the cables, related to their reference test methods are detailed.

This sectional specification is to be read in conjunction with EN 50288-1, which contains the essential provisions for its application.

Cables covered by this specification have maximum rated voltages of 90 V, 300 V and 500 V a.c.

These cables shall not be connected directly to mains electricity supply or other low impedance sources. Multi-element cables for use in analogue, digital and control circuits are not designed to be used for power supply.

These cables should be installed in accordance with the applicable local and national regulations.

Cables intended to have limited circuit integrity in a fire are not covered by this specification, but they are however under consideration for future editions.

There may be occasions when cables are required to have higher operating temperature ratings than those provided by using materials specified by the EN 50290 series. Suitable alternative materials are under consideration.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

<u>Publication</u>	<u>Year</u>	<u>Title</u>
EN 10218-1	-	Steel wire and wire products – General – Part 1: Test methods
EN 10244-2	-	Steel wire and wire products – Non-ferrous metallic coatings on steel wire – Part 2: Zinc or zinc alloy coatings
EN 10257-1	-	Zinc or zinc alloy coated non-alloy steel wire for armouring either power cables or telecommunications cables – Part 1: Land cables
EN 50289	Series	Communication cables - Specifications for test methods
EN 50290	Series	Communication cables
EN 50307	-	Lead and lead alloy sheath and sleeves of electric cables

EN 60708	-	Low-frequency cables with polyolefin insulation and moisture barrier polyolefin sheath (IEC 60708)
EN 60811-1-1	-	Insulating and sheathing materials of electric and optical cables – Common test methods – Part 1-1: General application – Measurement of thickness and overall dimensions – Tests for determining the mechanical properties (IEC 60811-1-1)
HD 383 S2	-	Conductors of insulated cables First supplement: Guide to the dimensional limits of circular conductors (IEC 60228 + IEC 60228A, mod.)
HD 446.3 S1	-	Thermocouples - Part 3: Extension and compensating cables – tolerances and identification system (IEC 60584-3, mod.)
IEC 60189-2	-	Low-frequency cables and wires with PVC insulation and PVC sheath - Part 2: Cables in pairs, triples, quads and quintuples for inside installations

3 Definitions, symbols and abbreviations

For the purposes of this European Standard the definitions of EN 50288-1 apply in addition to the following:

3.1

foil

a metal tape laminated to a plastic tape

4 Cable construction

4.1 Conductor

Except for thermocouple extension and compensating cables, with conductors as described in Annex D, conductors shall be solid, stranded or flexible plain or metal coated copper in accordance with Class 1, 2 or 5 of HD 383 in the range of 0,5 mm² to 2,5 mm². For multi-core cables the maximum conductor resistance shall be as HD 383, and for finished multi-pair, multi-triple and multi-quad cables the maximum resistance of HD 383 shall be increased by 2 %.

Conductor joints shall be as EN 50288-1.

Stranded and flexible conductors shall consist of wires circular in cross section assembled, without insulation between them, by concentric stranding or by bunching.

When the installed length of cable results in a high conductor resistance, larger conductor sizes can be used.

4.2 Insulation

The insulating material shall be selected from those listed below. It shall be applied to comply with the requirements of EN 50288-1.

- | | | |
|----|---------------------------------------|---------------|
| a) | PVC | EN 50290-2-21 |
| b) | Polyethylene | EN 50290-2-23 |
| c) | Polypropylene | EN 50290-2-25 |
| d) | Halogen free flame retardant compound | EN 50290-2-26 |
| e) | Cross-linked polyethylen | EN 50290-2-29 |

The minimum thickness at any point of the insulation shall not be less than the value given in Table 1, for the specified voltage rating, and the concentricity shall not fall below 75 % when measured using the procedure in Annex A.

There may be occasions when cables are required to have higher operating temperature ratings than those provided by using materials specified by the EN 50290 series. Suitable alternative materials are under consideration.

Table 1 – Minimum insulation thickness

Conductor size mm ²	Minimum thickness mm		
	Voltage rating		
	90 V	300 V	500 V
0,5	0,20	0,26	0,44
0,75	0,20	0,26	0,44
1,0	0,26	0,26	0,44
1,5	0,30	0,35	0,44
2,5	-	-	0,53

4.3 Cable elements

Cable elements shall be in accordance with 4.3 of EN 50288-1.

The lay length of a pair, triple or quad shall not exceed 100 mm for cables with conductor cross-section $\leq 1,5 \text{ mm}^2$, nor 150 mm for cables with conductor cross-section $2,5 \text{ mm}^2$.

4.4 Identification of cabling elements

Unless otherwise specified e.g. by means of numbered cores or tapes, the coding for identification shall be as given in IEC 60189-2 or EN 60708, as appropriate. The colours shall meet the requirements of 4.4 of EN 50288-1.

Coloured or numbered non-hygroscopic binder tapes may be applied over screened cabling elements as identification.

4.5 Screening of cabling elements

When screening of individual cable elements is specified, it shall be selected from those listed below and applied in accordance with 4.5 of EN 50288-1:

- a) a plain or coated metal braid with a minimum filling factor of 0,6 when calculated in accordance with Annex B;
- b) a combination of a foil, and a plain or coated metal braid with a minimum filling factor of 0,3 when calculated in accordance with Annex B. The use of a drain wire is optional when this type of screen is applied;
- c) a foil applied with a minimum overlap of 20 % and with a drain wire in direct contact with the metallic side of the foil.

Metal braid wire diameters are given in Table B.1.

4.6 Cable make-up

The cable elements shall be assembled together in concentric layers or in unit construction to form the cable core.

The cable core assembly shall be protected by wrappings of non-hygroscopic tape.

Successive layers may be separated from each other by interlayer binders of non-hygroscopic tape.

4.7 Filling compound

When a filling compound is specified, it shall be selected from the compounds given in EN 50290-2-28 and shall meet the requirements of 4.7 of EN 50288-1.

4.8 Interstitial fillers

When fillers are used they shall meet the requirements of 4.8 of EN 50288-1.

4.9 Screening of the cable core

The cable core shall be protected with a protective layer as described in 4.11 or 4.13 when it is covered with a screen.

When screening of the cable core is specified, it shall be selected from those listed below, and applied in accordance with 4.9 of EN 50288-1:

- a) a plain or coated metal braid with a minimum filling factor of 0,6 when calculated in accordance with Annex B;
- b) a combination of a foil, and a plain or coated metal braid with a minimum filling factor of 0,3 when calculated in accordance with Annex B. The use of a drain wire is optional when this type of screen is applied;
- c) a foil applied with a minimum overlap of 20 % and with a drain wire in direct contact with the metallic side of the foil.

Metal braid wire diameters are given in Table B.1.

Screening over the cable core may also be in the form of a laminated sheath (moisture barrier as 4.10 or multi layer sheath as 4.18.2).

4.10 Moisture barriers

When a moisture barrier is specified it shall be selected from those listed below. It shall be applied to meet the requirements of 4.10 of EN 50288-1:

- a) water swellable tapes;
- b) water swellable non-toxic powder;
- c) filling compound;
- d) laminated sheath, consisting of a longitudinal overlapped metallic foil, bonded within the overlapping and to the inner surface of an extruded sheath. The foil shall be one or both sides plastic coated, but in the case of only one side being coated a tinned copper drain wire shall be applied in direct contact with the metallic surface of the foil. The thickness of the metallic part of the foil shall be at least 0,15 mm;
- e) combinations of these four methods.

4.11 Protective wrappings

A wrapping layer may be applied under a sheath.

A wrapping layer may be applied under a metallic protection layer.

The wrapping layer shall be of a material compatible with the cable components with which it is in contact.

4.12 Inner sheath

When an additional inner sheath is applied under a metallic sheath (4.14) and may be applied under a lead sheath (4.18), it shall be selected from the materials listed below. It shall comply with the requirements of EN 50288-1:

- a) PVC to EN 50290-2-22;
- b) Polyethylene to EN 50290-2-24;
- c) Halogen free flame retardant compound to EN 50290-2-27.

The nominal thickness shall be calculated in accordance with Table C.1.

The nominal thickness (IS_{Rt}) shall be calculated like the nominal thickness (S_{Rt}) of the outer sheath without metallic protection (4.16).

A halogen free sheathing compound shall be used where the cores are insulated with a halogen-free compound.

There may be occasions when cables are required to have higher operating temperature ratings than those provided by using materials specified by the EN 50290 series. Suitable alternative materials are under consideration.

4.13 Bedding layers

A bedding layer shall be applied between a lead sheath and a metallic protection. This layer may be extruded (materials as in 4.12) or comprise helically applied tape or tapes.

The bedding layer shall be extruded when gland diameter requirements are specified.

The nominal thickness shall be calculated in accordance with Table C.5, where D is the diameter under the bedding layer calculated in accordance with Table C.1.

There may be occasions when cables are required to have higher operating temperature ratings than those provided by using materials specified by the EN 50290 series. Suitable alternative materials are under consideration.

4.14 Metallic protection

Metallic protection, when specified, shall be in accordance with 4.14 of EN 50288-1.

When round or flat galvanised steel wires are used, the minimum thickness shall be 0,80 mm (see Annex C). Flat galvanised steel wire may only be applied to cables where the diameter under the armour, calculated in accordance with Annex C, is > 15 mm.

The round steel wires of single wire armour shall comply with the requirements of EN 10257-1 and EN 10218-1. The coating of the wires shall comply with the requirements of EN 10244-2.

When a single or double layer of steel or brass tapes is used, the minimum thickness of tape shall be 0,20 mm for steel and 0,075 mm for brass.

When a metal braid is specified the minimum wire diameter shall be 0,30 mm. The minimum filling factor shall be 0,57 when calculated as Annex B.

Metal braid wire diameters are given in Annex C.

4.15 Integral suspension strand

Not applicable.

4.16 Outer sheathing

The outer sheathing material shall be appropriate to the operating environment and may be selected from the materials listed below. It shall be applied in accordance with 4.16 of EN 50288-1:

- a) PVC to EN 50290-2-22;
- b) Polyethylene to EN 50290-2-24;
- c) Halogen free flame retardant compound to EN 50290-2-27.

The nominal thickness (S_{Rt}) shall be calculated in accordance with the following formulae:

- a) without metallic protection $S_{Rt} = 0,04 \times D + 0,7 \text{ mm}$ (min 0,8 mm)
- b) with metallic protection $S_{Rt} = 0,028 \times D + 1,1 \text{ mm}$ (min 1,3 mm)

where D is the diameter under the outer sheath calculated in accordance with Table C.1.

For cables without metallic protection, the minimum thickness at a point shall not fall below the nominal value by more than 15 % + 0,1 mm. The minimum average thickness, rounded to 0,1 mm, shall not be less than nominal thickness.

For cables with metallic protection (4.14), the minimum thickness at any point shall not fall below the nominal value by more than 20 % + 0,2 mm

4.17 Fauna protection

When fauna protection is required, the method used shall be chosen from those in 4.17 of EN 50288-1.

There are no dimensional requirements.

4.18 Chemical and /or environmental protection

When chemical or environmental protection is required, one of the following methods shall be used.

4.18.1 Lead sheath

The lead sheath shall comply with EN 50307.

The nominal thickness (L_{Rt}) of the lead or lead alloy sheath shall be calculated using the following formula:-

$$L_{Rt} = 0,03 \times D + 0,7 \text{ mm (min of 0,8 mm)}$$

where D is the diameter under the lead sheath, calculated in accordance with Table C.1.

The minimum thickness at any point shall not be less than $0,95 L_{Rt} - 0,1 \text{ mm}$.

4.18.2 Multi layer sheath

The multi layer sheaths shall comprise a laminated sheath (see 4.10d) and an additional layer of polyamide.

The laminated sheath shall consist of a longitudinal overlapped aluminium foil with an extruded layer of high density polyethylene according to EN 50290-2-24. The thickness of the polyethylene layer shall be calculated as in 4.12 with a minimum thickness of 1,0 mm.

The thickness of the aluminium foil shall be at least 0,15 mm without coating. The coating may be on one or both sides.

The thickness of the polyamide covering shall be at least 0,3 mm at any point irrespective of cable diameter.

NOTE A material specification for polyamide is under consideration.

5 Test methods and requirements for completed cables

The following tables give the tests to be applied, together with the respective limits, in order to demonstrate compliance with this specification.

5.1 Electrical tests

5.1.1 Low-frequency and d.c. electrical measurements

EN 50288-1 Clause No.	Parameter	Test method	Requirement	
5.1.1.1	Conductor resistance	EN 50289-1-2	HD 383 for multicore cables and maximum shall be increased by 2 % for multi-pair, multi-triple, and multi-quad cables.	
5.1.1.2	Conductor resistance unbalance	EN 50289-1-2	N/A	
5.1.1.3	Dielectric strength	EN 50289-1-3	Duration 1 minute. For 90 V rating $\geq 0,75 \text{ kV}_{\text{ac}}$ or $\geq 1,5 \text{ kV}_{\text{dc}}$ For 300 V rating $\geq 1,0 \text{ kV}_{\text{ac}}$ or $\geq 2,0 \text{ kV}_{\text{dc}}$ For 500 V rating $\geq 2,0 \text{ kV}_{\text{ac}}$ or $\geq 3,0 \text{ kV}_{\text{dc}}$	
5.1.1.4	Insulation resistance	EN 50289-1-4	Material	Resistance $\text{M}\Omega/\text{km}$
			PVC	10
			Polyethylene	1 000
			Polypropylene	1 000
			HFFR	10
			XPPE	1 000
5.1.1.5	Mutual capacitance	EN 50289-1-5	Polyolefin	$< 150 \text{ nF/km}$
			Others	$< 250 \text{ nF/km}$
5.1.1.6	Capacitance Unbalance (pairs/quads)	EN 50289-1-5	Polyolefin	500 pF/500 m
5.1.1.7	Inductance	EN 50289-1-12	Only to be used for L/R	
5.1.1.8	Inductance to resistance ratio (L/R)	EN 50289-1-12 EN 50289-1-2	$< 25 \mu\text{H}/\Omega$ for up to 1 mm^2 $< 40 \mu\text{H}/\Omega$ for $1,5 \text{ mm}^2$ $< 60 \mu\text{H}/\Omega$ for $2,5 \text{ mm}^2$	

5.2 Mechanical tests

EN 50288-1 Clause No	Parameter	Test method	Requirement
5.2.1	Conductor elongation at break	EN 50289-3-2	$> 10 \%$
5.2.2	Shrinkage of insulation	EN 50289-3-4	EN 50288-1
5.2.3	Crush resistance of the cable	EN 50289-3-5	N/A
5.2.4	Impact resistance of the cable	EN 50289-3-6	N/A
5.2.5	Abrasion resistance of the sheath marking	EN 50289-3-8	N/A
5.2.6	Simulated installation testing of the cable	EN 50289-3-7	N/A
5.2.7	Tensile performance of the cable	EN 50289-3-16	N/A

5.3 Environmental tests

For further study (ffs).

5.4 Fire performance test methods

Fire performance test methods shall be in accordance with 5.4 of EN 50288-1.

Annex A (normative)

Insulation or sheath thickness and concentricity - Selection of samples and evaluation of results

A.1 Procedure for the measurement of thickness

The thickness of an insulation or sheath shall be measured in accordance with 8.1 and 8.5 of EN 60811-1-1. Three samples shall be taken from the cable at locations separated from each other by a distance of at least 1 m.

Compliance shall be checked on 10 % of all cores of a cable with at least at a minimum all cores of one cabling element.

If withdrawal of the conductor is difficult, it shall be stretched in a tensile machine or the sample shall be loosened by stretching or some other suitable means that does not damage the insulation.

A.2 Evaluation of results

The mean of the thickness of an insulation or sheath is the mean of six measurements from each sample expressed in mm. Measurements and calculations shall be to three decimal places and the result rounded to two decimal places. When the third decimal number is 5 or more, the result is rounded up and when it is 4 or less, the result is rounded down.

The minimum thickness of insulation at any point is the lowest value of all measurements.

A.3 Internal concentricity

The internal concentricity is the ratio of the smallest measured thickness to the largest measured thickness.

Annex B (normative)

Copper braid screening

The screening braid shall consist of tinned, annealed, high conductivity copper wires.

The number of spindles, ends per spindle and the diameter of the braid wires shall be sufficient to ensure that the filling factor (F) is not less than that specified. The filling factor shall be calculated from the following formula:

$$F = \frac{mnd_w}{2\pi D} \left(1 + \frac{\pi^2 D^2}{L^2} \right)^{1/2}$$

where

F is the filling factor

D is the mean diameter under the braid (mm)

d_w is the diameter of the braid wire (mm)

L is the lay length of the application of the braid (mm)

m is the total number of spindles

n is the total number of ends per spindle

Optical coverage (expressed as a percentage) = $(2F - F^2) \%$

Preferred braid wire diameters are shown in Table B.1.

The diameter of the wire shall be checked using a micrometer.

Table B.1 – Braid wire diameter

Mean diameter under the braid		Nominal diameter of braid wire	Minimum diameter of braid wire
Above	Up to and including		
mm	mm	mm	mm
-	3	0,10	0,098
3	6	0,15	0,146
6	15	0,20	0,196
15	25	0,30	0,296
25	-	0,40	0,396

Annex C (normative)

Determination of cable dimensions

C.1 Determination of cable dimensions

Starting with the conductor diameter and following the steps listed in Table C.1, the fictitious diameter F_D is increased progressively. Steps not applicable to a particular cable construction will give a zero increase in fictitious diameter.

Table C.1 – Steps to determine fictitious cable diameter

Step	Construction feature	Symbol	Determination
1	Conductor diameter	F_{CD}	Table C.2
2	Insulation radial thickness	I_{Rt}	Subclause 4.2 Table 1
3	Core element diameter	F_C	$F_{CD} + (2 \times I_{Rt})$
4	Pair element diameter	F_P	$2 \times F_C$
5	Triple element diameter	F_T	$2,16 \times F_C$
6	Quad element diameter	F_Q	$2,42 \times F_C$
7	Screened element diameter	$F_S^{2)}$	$F_E^{1)} + 0,3$
8	Additional separator tape (more than one screened element only)	F_{ESM}	$F_{ES} + 0,1$
9	Lay-up factors	L_F	Table C.3
10	Crush factors (more than one element only)	C_F	Table C.4
11	Cable make-up diameter	$F_A^{4)}$	$F_{CE}^{3)} \times L_F \times C_F$
12	Tape over lay-up	F_T	$F_A + 0,4$
13	Overall screen	F_{OS}	$F_T + 0,1$
14	Inner sheath thickness under lead sheath or metallic protection	I_{SRt}	Subclause 4.12
15	Diameter over inner sheath	F_{IS}	$F_{UIS}^{5)} + 2 \times I_{SRt})$
16	Lead sheath radial thickness	L_{Rt}	Subclause 4.18.1
17	Diameter over lead sheath	F_L	$F_{UL}^{6)} + 2 \times L_{Rt}$
18	Bedding radial thickness	B_{Rt}	Table C.5
19	Diameter over bedding	F_B	$F_L + 2 \times B_{Rt}$
20	Armour size (thickness or diameter) Round or flat wire Tape Braid	A_W	Subclause 4.14 and Table C.6 Table C.7 Table C.8
21	Diameter over armour	F_Z	$F_{UZ}^{7)} + K^{8)} \times A_W$
22	Outer sheath radial thickness	S_{Rt}	Subclause 4.16
23	Outer sheath diameter	F_D	Add $(2 \times S_{Rt})$

Notes:

- 1) F_E : Diameter of unscreened elements: F_C , F_P , F_T or F_Q
- 2) F_{ES} : Diameter of screened elements: F_{CS} , F_{PS} , F_{TS} or F_{QS}
- 3) F_{CE} : Diameter of cabling elements: F_E or F_{ESM}
- 4) F_A : For cabling element Core: $F_A = F_{CE} \times L_F$ (Crush factor has not to be applied)
- 5) F_{UIS} : Diameter under inner sheath = F_T or F_{OS}
- 6) F_{UL} : F_T or F_{OS} or F_{IS}
- 7) F_{UZ} : Diameter under armour = F_{IS} or F_B
- 8) K :
 - = 2 for flat or round wire armour
 - = 4 for double tape armour
 - = 4 for braid armour

C.2 Conductor diameters**Table C.2 – Conductor diameters**

Conductor cross section mm ²	Conductor diameter mm
0,50	0,80
0,75	1,00
1,00	1,10
1,50	1,40
2,50	1,80

C.3 Lay-up factors

Table C.3 – Lay up factors

No.	Fact	No.	Fact	No.	Fact	No.	Fact	No.	Fact
1	1,00	21	5,33	41	7,67	61	9,00	81	10,42
2	2,00	22	5,67	42	7,67	62	9,00	82	10,42
3	2,16	23	5,67	43	7,67	63	9,35	83	10,60
4	2,42	24	6,00	44	8,00	64	9,35	84	10,70
5	2,70	25	6,00	45	8,00	65	9,35	85	10,70
6	3,00	26	6,00	46	8,00	66	9,35	86	10,70
7	3,00	27	6,15	47	8,00	67	10,00	87	10,70
8	3,45	28	6,41	48	8,15	68	10,00	88	11,00
9	3,80	29	6,41	49	8,35	69	10,00	89	11,00
10	4,00	30	6,41	50	8,35	70	10,00	90	11,00
11	4,00	31	6,70	51	8,41	71	10,00	91	11,00
12	4,16	32	6,70	52	8,41	72	10,00	92	11,00
13	4,41	33	6,70	53	8,41	73	10,16	93	11,00
14	4,41	34	7,00	54	8,60	74	10,16	94	11,35
15	4,70	35	7,00	55	8,70	75	10,16	95	11,35
16	4,70	36	7,00	56	8,70	76	10,16	96	11,35
17	5,00	37	7,00	57	8,70	77	10,35	97	11,35
18	5,00	38	7,33	58	9,00	78	10,42	98	12,00
19	5,00	39	7,33	59	9,00	79	10,42	99	12,00
20	5,33	40	7,33	60	9,00	80	10,42	100	12,00

C.4 Crush factors

Table C.4 – Crush factors

Cable component	Crush factor
Ind screen pairs	0,889
Un/Col screen pairs	0,820
Ind screen triples	0,939
Un/Col screen triples	0,865
Cores/Quads	1,000

C.5 Bedding radial thickness

Table C.5 – Bedding radial thickness

Diameter under bedding		Bedding thickness
Above	Up to and including	
mm	mm	mm
-	25	1,0
25	35	1,2
35	45	1,4
45	60	1,6
60	80	1,8
80	-	2,0

C.6 Round wire armour

Table C.6 – Round armour wire

Diameter under armour		Wire diameter
Above	Up to and including	
mm	mm	mm
0	15	0,9 ± 0,035
15	25	1,25 ± 0,040
25	35	1,6 ± 0,045
35	45	2,0 ± 0,050
45	60	2,5 ± 0,060
60	-	3,15 ± 0,070

C.7 Tape armour

Table C.7 – Tape armour

Diameter under armour		Tape thickness
Above	Up to and including	
mm	mm	mm
-	30	0,2
30	70	0,5
70	-	0,8

C.8 Braid armour

Table C.8 – Braid armour wire size

Diameter under armour		Wire diameter
Above	Up to and including	
mm	mm	mm
-	20	0,3
20	-	0,4

Annex D (informative)

Thermocouple extension and compensating conductor combinations

D.1 Extension cables

Table D.1 – Extension cables

Combination code	Positive	Negative	Nominal e.m.f. (microvolts 0 °C/100 °C)
KX	Nickel chromium	Nickel aluminium	4,10
NX	Nickel chromium silicon	Nickel silicon	2,77
EX	Nickel chromium	Copper nickel (constantan)	6,32
JX	Iron	Copper nickel (constantan)	5,27
TX	Copper	Copper nickel (constantan)	4,28

D.2 Compensating cables

Table D.2 – Compensating cables

Combination code	Positive	Negative	Nominal e.m.f. (microvolts 0°C/100°C)
KCA	Iron	Copper nickel alloy	4,10
KCB	Copper	Copper nickel (constantan)	4,10
RCA	Copper	Copper low nickel alloy	0,65
RCB	Copper	Copper nickel mo alloy	0,65
SCA	Copper	Copper low nickel alloy	0,65
SCB	Copper	Copper nickel mo alloy	0,65
BC	Copper	Copper	0,03
NC	Copper nickel mg	Copper nickel mg	2,77

D.3 E.M.F.

The tolerance on e.m.f. values for the metal combinations used shall be in accordance with HD 446.3 S1.

D.4 Colour codes

The insulation colour code used shall be in accordance with HD 446.3 S1, or as agreed between the customer and supplier.